

5-day Value-at-Risk Estimation for a European Call Option Using Delta-Normal Approximation and Importance Sampling

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Abstract

This work presents a rigorous framework for estimating the *Value-at-Risk* (VaR) of a long European call option using the *Delta-Normal Approximation*. The underlying asset is modeled as a geometric Brownian motion on a filtered probability space $(\Omega, \mathcal{F}, \{\mathcal{F}_t\}_{t \geq 0}, \mathbb{Q})$ under the risk-neutral measure $\mathbb{Q}$. The approach combines Black-Scholes option pricing [1], Delta computation [2], and Importance Sampling using a Girsanov change of measure to assess tail risk. The 5-day VaR is formally defined following Artzner et al. [3].

1 Introduction

Value-at-Risk (VaR) quantifies the minimal loss a portfolio may incur over a given time horizon at a specified confidence level [3]. Let $(\Omega, \mathcal{F}, \{\mathcal{F}_t\}, \mathbb{P})$ be a filtered probability space supporting a standard Brownian motion. Then, for portfolio loss L over horizon Δt , the α -level VaR is defined as:

$$\text{VaR}_\alpha = \begin{cases} \inf\{x \in \mathbb{R} : \mathbb{P}(L > x) \leq \alpha\}, \\ L = V_0 - V_{\Delta t} \end{cases}$$

This work focuses on a long European call option, estimating the 5-day VaR via the Delta-Normal approximation. Importance Sampling is applied to better capture tail events [2], [4].

2 Theoretical Framework

2.1 Asset Dynamics

The underlying risky asset S_t^1 evolves according to the Black-Scholes model [1], [5]:

$$\begin{cases} dS_t^1 = \mu S_t^1 dt + \sigma S_t^1 dB_t, \\ S_0^1 > 0 \end{cases}$$

where μ is the drift, σ is volatility, and B_t is a standard Brownian motion adapted to the filtration $\{\mathcal{F}_t\}$. The risk-free bank account evolves as:

$$\begin{cases} dS_t^0 = r S_t^0 dt \\ S_0^0 = 1. \end{cases}$$

2.2 European Call Option Pricing

Let $C(t, F)$ denote the price of a European call option with strike K and maturity T . It satisfies the Black-Scholes PDE:

$$\begin{cases} \frac{\partial C}{\partial t} + \frac{1}{2}\sigma^2 F^2 \frac{\partial^2 C}{\partial F^2} - rC = 0, & 0 \leq t < T, \\ C(T, F(T, T)) = \max(F(T, T) - K, 0) \end{cases}$$

The analytical solution is:

$$C(t, S_t) = S_t N(d_1) - K e^{-r(T-t)} N(d_2),$$

with

$$\begin{cases} d_1 = \frac{\ln(S_t/K) + (r + \frac{1}{2}\sigma^2)(T-t)}{\sigma\sqrt{T-t}}, \\ d_2 = d_1 - \sigma\sqrt{T-t}. \end{cases}$$

2.3 Delta and Delta-Normal Approximation for VaR

The *Delta* of an option quantifies the sensitivity of the option's value to small changes in the underlying asset price. For a European call option, the Delta is given by

$$\Delta = \frac{\partial C}{\partial S_0^1} = e^{-rT} \mathbb{E}^{\mathbb{Q}} \left[\mathbf{1}_{\{S_T^1 > K\}} \frac{S_T^1}{S_0^1} \right] [2]$$

where the expectation is taken under the risk-neutral measure $\mathbb{Q}$. This expression captures the probabilistic impact of the terminal payoff on the option's current value, reflecting the likelihood that the option finishes in-the-money.

Under the *Delta-Normal Approximation*, for sufficiently small movements in the underlying asset, the change in the option value ΔV can be linearized as:

$$\begin{cases} \Delta V \approx \Delta \cdot V_0(S_0) \cdot \Delta S, \\ \Delta S \sim N(0, \sigma^2 \Delta t). \end{cases}$$

where ΔS represents the change in the underlying asset price over the risk horizon Δt , assumed to be normally distributed. This simplification allows us to model the portfolio loss using a univariate normal distribution, making the calculation of Value-at-Risk computationally efficient.

Consequently, the α -level Value-at-Risk over the horizon Δt is computed as

$$\begin{cases} \text{VaR}_\alpha = -z_\alpha \Delta \sigma \sqrt{\Delta t}, \\ z_\alpha = N^{-1}(1 - \alpha). \end{cases} [3]$$

where z_α is the α -quantile of the standard normal distribution. In this context, VaR represents the **minimum loss that will not be exceeded with probability $1 - \alpha$ **, providing a coherent measure of tail risk for risk management purposes.

2.4 Importance Sampling via Girsanov

To capture tail events more effectively, we shift the drift under a new measure $\mathbb{Q}^{\text{IS}}$ [2]:

$$\mu_{\text{IS}} = r - 0.5\sigma^2 + \text{shift factor} \cdot \sigma. \quad (1)$$

The Radon-Nikodym derivative gives the path weights:

$$w_i = \frac{d\mathbb{Q}}{d\mathbb{Q}^{\text{IS}}} = \exp(-(\mu_{\text{IS}} - r)\Delta t). \quad (2)$$

3 Algorithms

Algorithm 1 Delta-Normal VaR Estimation

- 1: **Input:** $S_0, K, r, \sigma, T, \Delta t, \alpha$
 - 2: Compute option Delta Δ
 - 3: Compute $z_\alpha = \text{qnorm}(1 - \alpha)$
 - 4: Compute VaR: $\text{VaR} = -z_\alpha \Delta \sigma \sqrt{\Delta t}$
 - 5: **Output:** VaR
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Algorithm 2 Importance Sampling for VaR Breaks

- 1: **Input:** $S_0, K, r, \sigma, T, \Delta t$, shift factor, N
 - 2: Initialize $Breaks = 0$
 - 3: **for** $i = 1$ to N **do**
 - 4: Simulate $S_i^{\text{IS}} = S_0 \exp((\mu_{\text{IS}} - 0.5\sigma^2)\Delta t + \sigma\sqrt{\Delta t}Z_i)$
 - 5: Compute option value V_i
 - 6: Compute weight $w_i = \exp(-(\mu_{\text{IS}} - r)\Delta t)$
 - 7: **if** $V_i - V_0 < \text{VaR}$ **then**
 - 8: $Breaks \leftarrow Breaks + w_i$
 - 9: **end if**
 - 10: **end for**
 - 11: Estimate $\hat{\alpha}_{\text{IS}} = Breaks/N$
 - 12: **Output:** $\hat{\alpha}_{\text{IS}}$
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4 Numerical Example

Parameter	Value
S_0	100
K	100
r	0.01
σ	0.5
T	1 year
Δt	5/365
α	0.05
N	10^6

Table 1: Simulation parameters

4.1 Results

The 5-day Value-at-Risk (VaR) for a long European call option, calculated using the Delta-Normal Approximation, is:

$$\text{VaR}_{5\text{-day}} = -5.837263.$$

This indicates that, under the model assumptions, the minimum expected loss in the worst 5% of cases over a 5-day horizon is approximately \$5.84. In other words, the portfolio is expected to lose at least this amount with 5% probability [3].

By incorporating Importance Sampling to better capture the tail of the distribution of asset returns, the estimated probability of observing a loss exceeding this VaR threshold is:

$$\hat{\alpha}_{\text{IS}} = 0.02256787.$$

This implies that in the simulated scenarios, only 2.26% of outcomes exceeded the computed VaR, which is lower than the nominal 5% level, reflecting the effectiveness of tail-focused sampling in capturing extreme events. These results highlight that using Importance Sampling can provide a more accurate estimate of tail risk, and help risk managers make informed decisions about capital allocation and potential hedging strategies.

Overall, the combination of the Delta-Normal approximation with Importance Sampling demonstrates a pragmatic balance between computational efficiency and the need for realistic tail-risk assessment, providing actionable insights for risk management in options portfolios.

5 Discussion and Improvements

- **Real-Time Data Integration:** Implementing live market data sources (e.g., via `yfinance`) allows continuous updating of VaR estimates to reflect current market conditions.
- **Model Enhancements:** The Delta-Normal Approximation assumes normally distributed returns, which underestimates extreme events. Incorporating stochastic volatility models (e.g., GARCH), fat-tailed distributions, or Monte Carlo simulations can provide more robust risk estimates.
- **Dynamic Risk Management:** VaR estimates can feed into dynamic hedging strategies, portfolio rebalancing, and risk limit frameworks to reduce the likelihood of large, unexpected losses.

6 Conclusion

The Delta-Normal Approximation offers a computationally efficient method to estimate VaR for European call options, capturing the primary sensitivity of the option to underlying price movements. Incorporating Importance Sampling via a Girsanov change of measure significantly improves the assessment of tail risk, providing a more accurate estimate of the probability and magnitude of extreme losses. Future work can enhance this framework by integrating real-time data, modeling non-normal return dynamics, and implementing fully dynamic risk management strategies to ensure robust portfolio protection under adverse market conditions.

References

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